



GROWTH & GH AXIS

HMG

HMG – 75 IU per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Human Menopausal Gonadotropin
SYNONYMS / ALIASES	hMG; menotropin; menotrophin
CAS NUMBER	9002-68-0
COMPOSITION	Mixture of follicle-stimulating hormone (FSH) and luteinizing hormone (LH) activity in approximately 1:1 ratio (typically 75 IU FSH + 75 IU LH per labeled "75 IU" dose)
FSH STRUCTURE	Heterodimeric glycoprotein; alpha-subunit (92 aa) + beta-subunit (111 aa); approximately 30 kDa glycosylated
LH STRUCTURE	Heterodimeric glycoprotein; alpha-subunit (92 aa) + beta-subunit (121 aa); approximately 28 kDa glycosylated
SOURCE	Purified from post-menopausal human female urine

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	Activity-based release (IU per vial for FSH and LH); chemical purity per batch COA
CONTAINER	75 IU per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	2-8 C (preferred for glycoprotein stability)
RECONSTITUTED STORAGE	2-8 C, use promptly; stability varies - confirm against COA
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	Typically 24 months at 2-8 C
STABILITY NOTES	Glycoprotein heterodimers are sensitive to elevated temperatures and freeze-thaw cycles. The FSH and LH components can degrade at different rates.
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STRUCTURAL NOTES

Human Menopausal Gonadotropin is a purified urinary preparation containing both follicle-stimulating hormone (FSH) and luteinizing hormone (LH) activity. Both FSH and LH are heterodimeric glycoproteins sharing a common alpha-subunit (92 amino acids, identical to the alpha-subunits of hCG and TSH) and possessing distinct beta-subunits (FSH-beta: 111 amino acids; LH-beta: 121 amino acids). The activity of each component is standardized in International Units (IU) against reference biological assays. Because hMG is a purified urinary product, residual urinary glycoproteins of post-menopausal origin may also be present at low levels.

SELECTED PEER-REVIEWED REFERENCES

1. Lunenfeld B. - *Historical perspectives in gonadotrophin therapy* - Human Reproduction Update, 2004 - DOI: 10.1093/humupd/dmh028
2. van Wely M. et al. - *Recombinant versus urinary gonadotrophin for ovarian stimulation in assisted reproductive technology cycles* - Cochrane Database of Systematic Reviews, 2011 - DOI: 10.1002/14651858.CD005354.pub2
3. Wide L., Bakos O. - *More basic forms of both human follicle-stimulating hormone and luteinizing hormone in serum at midcycle compared with the follicular or luteal phase* - Journal of Clinical Endocrinology & Metabolism, 1993 - DOI: 10.1210/jcem.76.4.8473403

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.